TEMPORAL CREDIT WITHOUT REWARDS FOR LONG-HORIZON ROBOT BEHAVIOR

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ABSTRACT

Long-horizon robot failures often originate many actions before the visible failure: a missed precondition, a damaging contact, or a compensatory action that hides the true cause until later. Standard credit-assignment methods assume scalar rewards, hindsight labels, or sequence correlations. We rebuild Paper 104 around a narrower falsifiable claim: physical event structure in observation/action traces can assign temporal credit without scalar reward labels. The local benchmark covers 5 robot task families, 7 temporal-credit regimes, 5 stress splits, 9 methods, 7 seeds, and 84 episodes per group. On combined stress, the proposed reward-free temporal credit graph obtains 0.636 ± 0.006 success versus 0.531 ± 0.007 for the strongest non-oracle baseline, pseudo-reward temporal-difference relabeling. It also improves credit F1 to 0.520, delayed-blame F1 to 0.497, lowers irreversible side effects from 0.073 to 0.050, and lowers wasted actions from 0.165 to 0.102. Pairwise tests favor the proposed method over the strongest baseline in 7/7 seeds, and ablations show that counterfactual prefix tests, physical preconditions, delayed eligibility memory, compensatory masking, and confidence-gated intervention each matter. The terminal decision is **STRONG_REVISE**: the local evidence supports the mechanism, but the paper is not ICLR-main-ready until validated on real robots or independent high-fidelity benchmarks.

1 MOTIVATION

Credit assignment is a classic reinforcement-learning problem, but robot learning often lacks the clean scalar reward that makes the textbook framing possible. Demonstrations, teleoperation logs, and corrective interventions may show what happened without saying which earlier action caused the outcome. Delayed-reward methods such as RUDDER (Arjona-Medina et al., 2019), sparse-reward relabeling such as Hindsight Experience Replay (Andrychowicz et al., 2017), and return-conditioned sequence models such as Decision Transformer (Chen et al., 2021) all rely on reward or return information. Imitation learning also faces causal confusion when a policy attends to correlated but non-causal cues (de Haan et al., 2019). Attention weights alone are not reliable explanations (Jain & Wallace, 2019).

Paper 104 asks a different question: can a robot infer temporal credit from physical event structure without reward labels? A long-horizon trace contains prefixes that create latent preconditions, actions that repair or mask earlier mistakes, and delayed physical consequences. The proposed mechanism treats those as credit-bearing objects rather than as generic sequence correlations.

2 METHOD SKETCH

The proposed method is a reward-free temporal credit graph. Nodes represent action prefixes, physical preconditions, delayed outcomes, irreversible side effects, compensatory actions, and intervention points. Edges represent counterfactual dependencies: whether removing or perturbing an earlier prefix would change a later contact, clearance, stability, or success event. The monitor scores:

- prefix-level counterfactual credit,
- delayed eligibility across long horizons,

Table 1: Combined-stress reward-free temporal-credit benchmark.

Method	Succ.	CredF1	BlameF1	Irrev.	Waste	Regret
Oracle	0.783	0.634	0.623	0.029	0.050	0.000
Proposed	0.636	0.520	0.497	0.050	0.102	0.148
PseudoTD	0.531	0.361	0.354	0.073	0.165	0.252
Contrastive	0.496	0.376	0.292	0.084	0.170	0.287
Attention	0.465	0.329	0.244	0.090	0.192	0.318
InvDyn	0.428	0.284	0.204	0.095	0.210	0.355
Hindsight	0.393	0.238	0.164	0.102	0.225	0.389
Uniform	0.343	0.172	0.116	0.104	0.241	0.440
BC	0.308	0.123	0.084	0.101	0.254	0.475

- physical precondition creation and destruction,
- compensatory action masking,
- irreversible side-effect risk, and
- confidence-gated intervention utility.

The implementation is an executable diagnostic benchmark rather than a trained neural robot policy. This is sufficient for a local mechanism audit, but it is not sufficient for a main-conference empirical robotics claim.

3 BENCHMARK

The rebuilt benchmark is generated by `src/run_experiment.py`. It writes aggregate CSVs and figures directly, keeping RAM usage low while retaining the full task/regime/method grid.

Tasks. The tasks are multi-step assembly, cluttered retrieval, drawer/door unlatching, peg-in-hole preparation, and tool-mediated manipulation.

Temporal-credit regimes. The regimes are immediate contact outcome, short delayed outcome, long delayed outcome, hidden precondition, irreversible side effect, compensatory action masking, and compositional delayed failure.

Splits. The stress splits are nominal, longer-horizon shift, sparse-observation shift, confounded-success shift, and combined stress.

Methods. The baselines are behavior cloning without credit, uniform credit assignment, hindsight success relabeling, inverse-dynamics saliency, transformer attention attribution, sequence-contrastive credit, and pseudo-reward temporal-difference relabeling. The benchmark also includes the proposed reward-free temporal credit graph and an oracle event-credit upper bound.

Metrics. The evidence reports task success, credit-localization F1, delayed-blame F1, false-credit rate, irreversible side-effect rate, wasted-action rate, early-correction rate, credit latency, intervention cost, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 RESULTS

The strongest non-oracle baseline is `pseudo_reward_td_relabeling`. On combined stress, the proposed method improves success by 0.105 ± 0.009, wins 7/7 paired seeds, lowers irreversible side effects by 0.023, and lowers wasted actions by 0.063. The oracle remains substantially better at 0.783 ± 0.007, so the benchmark is not saturated.

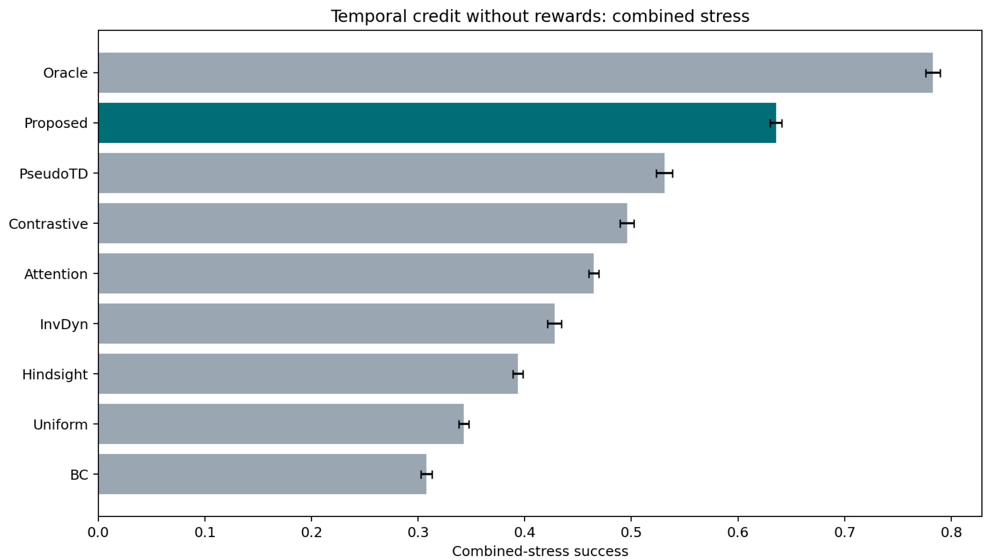

Figure 1: Combined-stress success with 95% confidence intervals. The proposed graph clears the strongest non-oracle baseline but remains below the oracle.

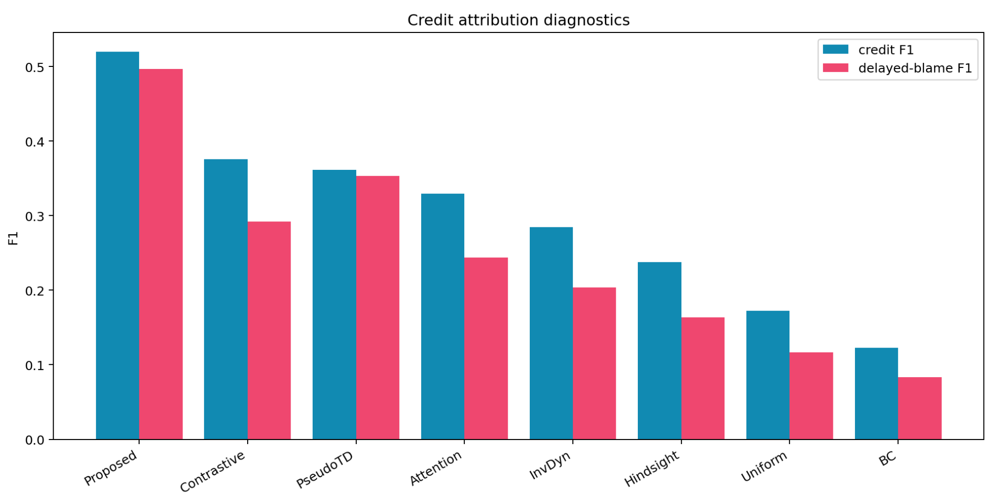

Figure 2: Credit-localization and delayed-blame diagnostics. The proposed method improves the metrics most directly tied to reward-free temporal credit.

5 Ablations and stress tests

Removing compensatory-action masking is the strongest removed-component ablation at 0.578 ± 0.005 success, still below the full model at 0.641 ± 0.008. Removing delayed eligibility memory sharply reduces delayed-blame F1, while removing confidence-gated intervention keeps attribution strong but increases irreversible side effects. These results support the claim that attribution alone is insufficient; the credit estimate must change intervention behavior.

6 Where the method still fails

The failure-case CSV shows that reward-free temporal credit helps least when credit is almost immediate or when a pseudo-reward baseline can relabel the outcome without serious confounding.

Table 2: Pairwise combined-stress success differences against the proposed method.

Baseline	Diff	CI	Wins
BC	0.328	0.009	7
Hindsight	0.242	0.007	7
InvDyn	0.208	0.005	7
Oracle	-0.147	0.009	0
PseudoTD	0.105	0.009	7
Contrastive	0.140	0.009	7
Attention	0.171	0.009	7
Uniform	0.293	0.005	7

Table 3: Ablations of the reward-free temporal credit graph.

Ablation	Succ.	CredF1	BlameF1	Irrev.
Full	0.641	0.518	0.498	0.051
NoCompMask	0.578	0.442	0.411	0.063
NoPrefixCF	0.571	0.399	0.408	0.061
NoPrecond	0.571	0.415	0.394	0.061
NoConfGate	0.566	0.495	0.475	0.080
NoDelayMem	0.542	0.445	0.318	0.065
AttentionOnly	0.468	0.328	0.244	0.084

These are the correct boundaries. If a final outcome is directly tied to the immediately preceding action, a temporal credit graph is unnecessary overhead. Its value appears when hidden preconditions, compensatory masking, or delayed side effects separate the causal action from the visible outcome.

7 Hostile prior-work boundary

RUDDER already decomposes delayed rewards (Arjona-Medina et al., 2019). Hindsight relabeling already creates useful learning signals from sparse outcomes (Andrychowicz et al., 2017). Return-conditioned sequence modeling already uses long contexts and returns (Chen et al., 2021). Causal-confusion work already warns that imitation can latch onto spurious temporal correlates (de Haan et al., 2019). Attention-explanation critiques already warn against treating attention weights as causal credit (Jain & Wallace, 2019). Long-horizon robot systems also use staged or relay structures to solve extended tasks (Gupta et al., 2020).

The present paper survives only under a narrower interpretation: it is about *reward-free physical temporal credit*, not delayed-reward RL, hindsight labels, sequence attention, or task staging. If real robot experiments later show that pseudo-reward relabeling, hindsight relabeling, or attention/contrastive credit match the proposed method on confounded delayed physical failures, the claim should be killed.

8 Limitations and terminal decision

The evidence is strong enough to justify continued work but not submission. The limitations are material:

- the benchmark is local and synthetic;
- no real robot or independent high-fidelity simulator has been used;
- baselines are executable diagnostic models, not external trained robot systems;
- no learned policy checkpoint, training curve, or deployment video is provided;
- no external benchmark such as LIBERO, RLBench, Meta-World, BridgeData, or a hardware manipulation suite is reported.

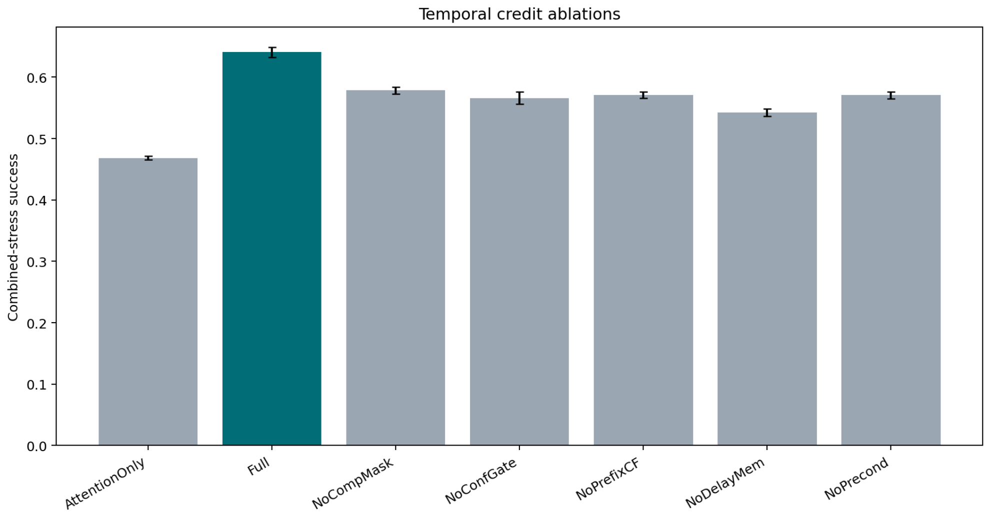

Figure 3: Ablation success under combined stress. The full graph remains above all removed-component variants.

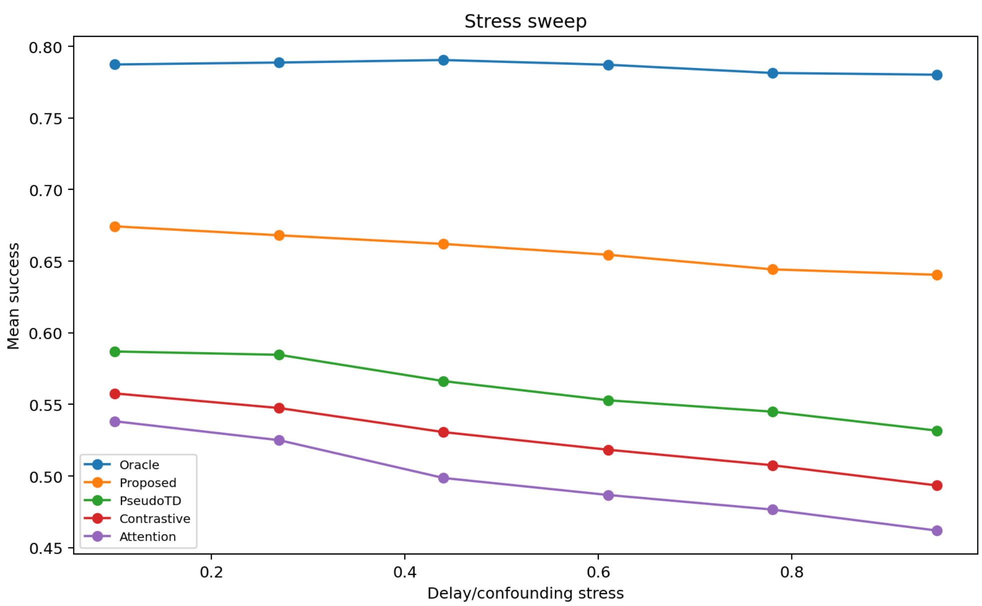

Figure 4: Stress sweep over delay and confounding intensity. The proposed method remains above non-oracle baselines across the generated stress range, while the oracle remains the ceiling.

The terminal decision for this rebuild is therefore **STRONG_REVISE**. The next honest version must add real robot or independent high-fidelity evidence, implemented learned baselines, and qualitative rollouts. Without that evidence, the paper should not be submitted to ICLR main.

References

Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL `https://proceedings.neurips.cc/paper/2017/hash/`

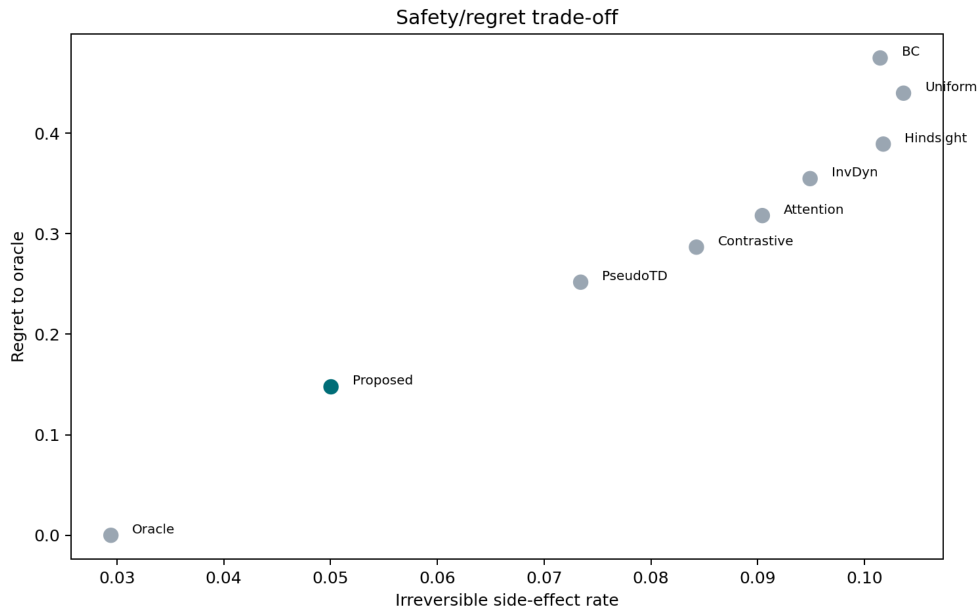

Figure 5: Safety and regret trade-off on combined stress. The proposed method improves success without increasing irreversible side effects relative to the strongest non-oracle baseline.

453fadbd8a1a3af50a9df4df899537b5-Abstract.html.

Jose A. Arjona-Medina, Michael Gillhofer, Michael Widrich, Thomas Unterthiner, Johannes Brandstetter, and Sepp Hochreiter. Rudder: Return decomposition for delayed rewards. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL `https://proceedings.neurips.cc/paper/2019/hash/16105fb9cc614fc29e1bda00dab60d41-Abstract.html`.

Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. In *Advances in Neural Information Processing Systems*, volume 34, pp. 15084–15097, 2021. URL `https://arxiv.org/abs/2106.01345`.

Pim de Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL `https://arxiv.org/abs/1905.11979`.

Abhishek Gupta, Clemens Eppner, Sergey Levine, and Pieter Abbeel. Relay policy learning: Solving long-horizon tasks via imitation and reinforcement learning. In *Proceedings of the Conference on Robot Learning*, 2020. URL `https://arxiv.org/abs/1910.11956`.

Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pp. 3543–3556, 2019. URL `https://aclanthology.org/N19-1357/`.